Multi-Protocol IoT Development & Evaluation Kit

(Transparent Transmission Series)

Models Covered:

- **UART-WiFi-001** (UART-to-WiFi Evaluation Board)
- **SPI-WiFi-002** (SPI-to-WiFi Evaluation Board)
- **CAN-WiFi-003** (CAN-to-WiFi Evaluation Board)
- **RS485-WiFi-004** (RS485-to-WiFi Evaluation Board)

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1. Product Introduction

The **CONV-WiFi Series** is an engineering evaluation platform based on the **ESP32-C3-MINI-1-N4** module. It is designed as a bidirectional data pipeline to facilitate **Transparent Data Transmission** evaluation between field-side MCU interfaces (CAN, SPI, RS485, UART) and remote servers.

IMPORTANT NOTICE: This product is a **sub-assembly/development component** and is **NOT** a finished consumer product. It is intended solely for professional engineering development, demonstration, or evaluation purposes in a laboratory environment.

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2. Technical Specifications (Reference for Developers)

- **Core Controller:** ESP32-C3-MINI-1-N4 (RISC-V 32-bit MCU up to 160 MHz).
- **Power Supply Options:**
  - **5V Systems:** 4.5V to 5.5V DC via 5V pin or USB (onboard regulator to 3.3V).
  - **3.3V Systems:** 3.3V DC directly to 3V3 pin.
- **Current Requirement:** A stable source capable of providing at least **1.0 A** is recommended to support peak Wi-Fi and bus activity.
- **Safety Warning:** Use only the provided 5.0V power adapter (Model: VEL05US050-US-JA) when powering via USB. Do not exceed 5.5V.
- **Interface Capabilities:**
  - **UART:** 300–115200 bps, 8N1.
  - **SPI:** Slave mode only, Max speed 10 MHz.
  - **CAN 2.0B:** 25 kbps – 1 Mbps (External 120Ω termination required).
  - **RS485:** 300–115200 bps, 8N1.

- **Wireless:** IEEE 802.11b/g/n (2.4 GHz) with 1T1R mode up to 150 Mbps.
- **Operating Temp:** -40 °C to +85 °C / +105 °C.
- **Dimensions:** 31 mm × 40 mm × 10 mm (Base board).

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3. Hardware Connection Guidelines

- **Voltage Matching:** Select power input based on the target MCU's system voltage.
- **Grounding:** Ensure the **GND** pin is connected to the system's common ground.
- **Signal Integrity:** Use level shifters for 5V tolerance on UART/SPI as required.
- **Warning:** Hardware connections should be performed by qualified personnel familiar with embedded systems.

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4. FCC & ISED Compliance Statement

Note on Regulatory Status: This device is an **Evaluation Kit** and complies with Part 15 of the FCC Rules.

- **RF Module Information:** Contains ESP32-C3-MINI-1-N4 module with **FCC ID: 2AC7Z-ESP32C3MINI1**.
- **Compliance Boundary:** This sub-assembly is designed to meet the technical intent of **FCC Part 15 Subpart B Class B** and Canadian **ICES-003 Issue 7 Class B**.
- **Responsibility for Final Products:** * This kit is **NOT** a certified finished product.
    - The manufacturer (WNetTech) provides this board for development only.
    - **Final integrators** incorporating this kit into a commercial product are solely responsible for obtaining their own final-product certifications (e.g., FCC Part 15B / ICES-003).
- **RF Exposure:** Maintain a separation distance of at least **20 cm** from all persons during operation.

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5. Contact Information

- **Manufacturer:** WNetTech
- **Address:** 11 Charlemont Crescent, Scarborough, ON, M1T 1M3, Canada
- **Email:** kevinzhang61@hotmail.com

- **Compliance Documentation:** https://github.com/wnettech/compliance